SECTION 02 41 19

SELECTIVE DEMOLITION

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Selective Demolition for Remodeling / Renovations.

B. Salvage Items.

1.03 DEFINITIONS

A. Demolish: Complete demolition of an entire structure. No items to be salvaged. All demolished material shall be disposed of in a legally and appropriate manner off-site.

B. Remove: Selective removal of a designated building element in a manner that ensures that surrounding existing building elements (to remain) are protected from damage. All removed items shall be disposed of in a legally and appropriate manner off-site.

C. Salvage: Removal of a designated building element in a manner that protects the item being removed for re-use in the Project, or turned over to the Owner. Item is to be stored by the Contractor (for re-use) or coordinated with the Owner for removal from the site.

1.04 SCOPE OF SELECTIVE DEMOLITION WORK

A. Visit the site prior to bid and examine the existing conditions as to character and extent of work involved.

B. Removal of interior partitions and finishes.

C. Provide selective demolition work as follows:

1. As required to admit, accommodate and complete the work of this project shown on the Contract Documents;

2. As shown on the Drawings;

3. As specified herein;

4. Note: Demolition and removal work not specifically shown on the Drawings but that is required to complete the new Work shown is Contract Work included in the Base Bid and will not be considered as additional work.

D. Carefully remove finishes required to admit and allow new Work; coordinate cut lines and removal techniques with installer doing the finish material patching and repair work so as to minimize patching and repair work.

E. Protect existing building structure and finishes not scheduled for removal from damage during selective demolition.

F. Protect existing mechanical and electrical systems from damage or contamination with demolition dirt / dust or water during selective demolition.

1.05 REGULATORY REQUIREMENTS

A. Conform to applicable codes for demolition work, dust control, products requiring mechanical and electrical disconnection and re-connection.

B. Obtain required permits from authorities having jurisdiction.

C. Do not close or obstruct egress from any building exit or site exit.

D. Conform to applicable codes and ordinances for disposal of construction / demolition debris.

E. Conform to procedures and requirements of authorities having jurisdiction when hazardous or contaminated materials are discovered. Notify the Owner and Architect immediately.

1.06 PROJECT CONDITIONS / EXISTING BUILDING SYSTEMS

A. Conduct demolition to minimize interference with adjacent and occupied building areas.

B. Cease operations immediately if structure appears to be in danger and notify Architect. Do not resume operations until directed.

C. Provide, erect and maintain temporary barriers and security devices as necessary to protect the public.

1.07 HAZARDOUS MATERIALS

A. The Owner has not completed a hazardous materials survey of the existing building. In the event hazardous materials are encountered during demolition, notify the Owner and Architect and request direction.

PART 2 - PRODUCTS

2.01 DEMOLITION EQUIPMENT

A. Provide dumpsters or trucks to receive debris.

1. Do not throw debris onto the ground; take measures required to prevent debris from accidentally falling out of truck / dumpster.

B. Demolition equipment utilized inside building shall not exceed load capacity of existing floor structure; consult with Structural Engineer on existing structure load capacity prior to bringing any heavy equipment on site.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the overall Work of this project in a coordinated and efficient manner.

B. Coordinate demolition work with other trades to accommodate or admit the work of this Project.

3.02 SALVAGE ITEMS

A. Any existing item identified by Owner prior to removal from building / site shall be set aside and protected for salvage by Owner.

B. As noted on the Drawings, salvage the removed Granite Wall Panel Tiles for reuse.

3.03 PREPARATION

A. Inspect site and verify extent and location of items designated for removal, disposal, alternative disposal, recycling, salvage and items to remain.

B. Provide necessary temporary utility by-pass connections to facilitate installation of new work and minimize disruption to Owner’s occupancy of other buildings.

C. Provide bracing, shoring and temporary structures / supports required to prevent any movement of permanent partitions, structure or mechanical / electrical systems due to demolition or construction work.

D. Verify that existing conditions are consistent with the Drawings, if they are not, obtain approval from the Project structural engineer prior to demolition, saw-cutting or core drilling of any structural element or part of the building structure.

E. Lay out and coordinate the demolition work so that new work can be completed free from conflicts with work of other trades and existing conditions / systems not scheduled for removal. Do not proceed with demolition work until conflicts are resolved.

F. Provide appropriate temporary signage including signage for exit or building egress. Include warning signs to public at perimeter of construction limits to prohibit access to non-authorized personnel.

G. Dispose of offsite any salvage items that Owner does not want.

3.04 PROTECTION

A. Execute demolition work in an orderly and careful manner, with due consideration for any existing structures, including any parts of the surrounding areas which are to remain.

B. Protect adjacent construction to remain from dust and debris resulting from demolition activities.

C. Barricade, brace and cover as necessary to protect the public, workmen and occupants.

D. Wherever necessary for protection of persons or property, brace or shore walls, partitions, or structure as required.

E. Erect and maintain weatherproof, secure temporary closures for exterior openings.

F. Provide means to prevent dust from exiting building and construction areas or from entering occupied area within the building. Contain dust within each work area or room.

3.05 DEMOLITION

A. The general scope of demolition work includes, but is not limited to the following:

1. Selective demolition to admit, accommodate and complete the work of this project shown on the Contract Documents.

B. Provide the tools and equipment best suited for the specific type of demolition, cutting, moving or removing required.

C. Plan and execute demolition work in a manner that results in the least negative impact to the surrounding work.

D. Cut, move or remove items as shown on Drawings to provide access for renovation Work and as necessary to admit, conceal or complete the work of this contract.

E. Demolished or removed items and materials are the property of the Contractor, except salvage items claimed by the Owner or shown on the Drawings and shall be removed from the site and disposed of legally. Do not burn or bury materials on site.

F. Remove materials as the work progresses. Upon completion of work, leave areas in clean condition.

G. Remove temporary work.

H. Cease operations and notify Architect immediately if existing demolition items and / or components appear to contain hazardous materials. Promptly notify Architect. Do not resume operations until corrective measures have been taken.

I. Cease operations and notify Architect immediately if existing structural system and / or components appear to be endangered. Do not resume operations until corrective measures have been taken.

J. Remove and promptly dispose of contaminated, vermin infested or dangerous materials encountered.

3.06 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices.

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

END OF SECTION

SECTION 07 90 00

JOINT SEALERS

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Interior Joint Sealants.

1.03 REFERENCES

A. References shall be the edition current as of the date of the Contract Documents.

B. ASTM C834 - Standard Specification for Latex Sealants.

C. ASTM C1193 - Standard Guide for Use of Joint Sealants.

D. ASTM D1667 - Standard Specification for Flexible Cellular Materials—Poly (Vinyl Chloride) Foam (Closed-Cell).

1.04 SUBMITTALS

A. Refer to Division 01 for submittal procedures.

B. Product Data: Provide data indicating sealant chemical characteristics, performance criteria, substrate preparation, limitations, and color availability.

1.05 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the Products specified in this section with a minimum of five (5) years of experience.

B. Applicator Qualifications: Company specializing in performing the work of this section with a minimum of five (5) years of experience.

1.06 ENVIRONMENTAL REQUIREMENTS

A. Maintain temperature and humidity recommended by the sealant manufacturer during and after installation.

1.07 COORDINATION

A. Coordinate the work with all sections referencing this section.

PART 2 - PRODUCTS

2.01 INTERIOR SEALANTS

A. General Purpose Interior Sealant: Siliconized acrylic emulsion latex; ASTM C834, single component, paintable.

1. Color: Match color of adjacent materials; or as selected by Architect.

2. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. Pecora; *AC-20+Silicone.*

b. Tremco; *Tremflex 834.*

c. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified products performance and colors.

3. Applications: Use for:

a. Joints between interior relite frames and wall surfaces.

b. Between GWB and other materials.

c. Other interior joints for which no other type of sealant is indicated.

2.02 ACCESSORIES

A. Primer: Non-staining type, recommended by sealant manufacturer to suit application.

B. Joint Cleaner: Non-corrosive and non-staining type, recommended by sealant manufacturer; compatible with joint forming materials.

C. Joint Backing: Round foam rod compatible with sealant of type recommended by sealant manufacturer for type of sealant; ASTM D1667, oversized as recommended by sealant manufacturer.

D. Bond Breaker: Pressure sensitive tape recommended by sealant manufacturer to suit application.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the overall Work of this project in a coordinated and efficient manner.

3.02 EXAMINATION

A. Inspect the substrate surfaces and joint openings and confirm they are ready to receive sealant work.

B. Confirm that joint size, configuration and conditions conform to sealant manufacturer's requirements so as to achieve manufacturer's published sealant performance.

C. Verify that joint backing and release tapes are recommended for use by sealant manufacturer with the specified sealant.

D. Do not start sealant installation until substrate surfaces and joint openings conform to sealant manufacturer’s requirements.

E. Sealant Adhesion Testing: Test adhesion of the sealant to each different substrate material in accordance with manufacturer’s requirements; install primer recommended by sealant manufacturer if any adhesion test fails.

F. Start of sealant installation indicates installer’s acceptance and confirmation that substrate, joint openings and conditions are in conformance with sealant manufacturer’s requirements.

3.03 PREPARATION

A. Thoroughly clean and prepare joint substrate surfaces in accordance with sealant manufacturer's instructions to achieve published sealant performance.

B. Remove loose materials and foreign matter which might impair adhesion of sealant.

C. Clean and prime joints in accordance with manufacturer's instructions.

D. Protect elements surrounding the work of this section from damage or disfigurement.

3.04 INSTALLATION – SEALANTS, GENERAL

A. Perform work in accordance with sealant manufacturer's requirements for preparation of surfaces and material installation instructions to achieve published sealant performance.

B. Perform installation in accordance with ASTM C1193.

C. Measure joint dimensions and size joint backers to achieve width-to-depth ratio, neck dimension, and surface bond area as recommended by manufacturer.

D. Install bond breaker where joint backing is not used.

E. Install sealant free of air pockets, foreign embedded matter, ridges, and sags.

F. Apply sealant within recommended application temperature ranges. Consult manufacturer when sealant cannot be applied within these temperature ranges.

G. Tool joints concave.

3.05 CLEANING

A. Clean adjacent soiled surfaces.

3.06 PROTECTION OF FINISHED WORK

A. Protect sealants until cured.

3.07 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices.

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

END OF SECTION

SECTION 08 11 00

HOLLOW METAL FRAMES

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Hollow Metal Relite Frames.

1.03 REFERENCES

A. References shall be the edition current as of the date of the Contract Documents.

B. ANSI/SDI A250.8 - Specifications for Standard Steel Doors and Frames (SDI-100).

C. ANSI/SDI A250.10 - Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames.

D. ANSI/SDI A250.11 - Recommended Erection Instructions for Steel Frames.

E. ASTM A568/A568M - Standard Specification for Steel, Sheet, Carbon, Structural, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for.

F. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.

G. ASTM A924/A924M - Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process.

H. ASTM A1008/A1008M - Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable.

I. ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength.

J. ASTM C518 - Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus.

K. ASTM C591 - Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation.

L. ASTM C1363 - Standard Test Method for Thermal Performance of Building Materials and Envelope Assemblies by Means of a Hot Box Apparatus.

M. ASTM D3359 - Standard Test Methods for Rating Adhesion by Tape Test.

N. NFPA 80 - Standard for Fire Doors and Other Opening Protectives.

O. SDI 111-A - Standard Steel Door Frame Details.

P. WAC 51-50 - Washington Administrative Code, State of Washington Building Code.

1.04 SUBMITTALS

A. Refer to Division 01 for submittal procedures.

B. Product Data: Provide manufacturer's product literature and standard details.

1. Provide manufacturer’s technical data sheet on each different type of anchor and hardware reinforcement required.

C. Shop Drawings: Provide elevation and details of each different frame type, including frame anchors, glass stops, vision panels, and special conditions.

1.05 QUALITY ASSURANCE

A. Manufacturer Qualifications: Provide products from a single manufacturer who is a member of the Steel Door Institute.

B. Regulatory Requirements:

1. Code Compliance: Comply with WAC 51-50 amendments to the International Building Code.

1.06 DELIVERY, STORAGE AND HANDLING

A. Protect products from moisture, construction traffic, and damage.

B. Store vertically under cover. Do not use non-vented plastic or canvas shelters. Should wrappers become wet, remove immediately.

PART 2 - PRODUCTS

2.01 MANUFACTURERS

A. Manufacturers: Manufacturers shall be a member of Steel Door Institute with products conforming to the requirements of the Drawings and this specification.

2.02 MATERIALS

A. Steel Sheet for Frames:

1. Cold Rolled Steel: ASTM A1008/A1008M and ASTM A568/A568M.

2. Hot Rolled Steel: Pickled and oiled, ASTM A1011/A1011M and ASTM A568/A568M.

3. Galvanized Steel: ASTM A924/A924M and ASTM A653/A653M; hot-dipped zinc-coated steel.

B. Steel Sheet for Anchors and Accessories: ASTM A653/A653M, hot-dipped zinc-coated steel; with minimum G60 metallic coating.

2.03 HOLLOW METAL (HM) RELITE FRAMES

A. Frames: ANSI/SDI A250.8 Level 3 – Extra Heavy Duty is the minimum performance standard; provide the following special requirements that exceed this minimum standard:

1. Regular Use Frames: 16-gauge steel sheet, fabricated to size, profile and configuration shown on Drawings.

2. Fire-Rated Frames: Tested and listed by code approved testing agency for fire rating required for the wall that frame is mounted in.

a. Fire Label Requirement: Provide a fire label permanently affixed to the frame. Fire rating information can be a separate label, or it can be stamped / embossed into the frame.

3. Corner Construction: Face weld corners, grind welds flush and smooth.

4. Jamb Anchors: Provide jamb anchors per SDI 111-A.

a. Metal Framing: Sheet Metal 'Z' Clip Steel Stud Anchor, no wrap around nail on straps.

5. Glazed Lights in Frames: Provide rolled steel channel shape glazing stops prepared for countersunk oval head screws and butted corners.

a. Screws: Provide tamper-resistant screws.

1) Screw heads shall be painted to match the stops and frames.

6. Tamper-Resistant Screws: Provide oval head, center pin Torx drive screws.

2.04 FABRICATION

A. Confirm field conditions and coordinate depth of each frame throat to match thickness of wall or other configuration shown on Drawings.

B. Fabricate steel frames to sizes and profiles shown on the Drawings in conformance to the requirements of this section, ANSI/SDI A250.8 and fire listing requirements.

C. Finish:

1. Factory Prime Paint Finish: Prime paint surfaces of frames under controlled conditions at the factory.

a. Frames shall be thoroughly cleaned, and chemically treated to insure maximum paint adhesion.

b. Surfaces of the frames exposed to view shall receive a factory applied coat of rust inhibiting primer, either air-dried or baked-on.

c. The finish shall meet the requirements for acceptance stated in ANSI/SDI A250.10.

2. Shop Prime Paint Touch-Up: Repair any factory applied prime paint damaged by shipping or by shop modifications to frames.

a. Surface preparation, prime paint and application shall conform to factory finishing standards and be compatible with field painting specified in Section 09 90 00.

b. The finish shall meet the requirements for acceptance stated in ANSI/SDI A250.10.

3. Performance Requirement: Primer bond to steel substrate shall pass adhesion field testing per ASTM D3359, Type A Cross Hatch.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the overall Work of this project in a coordinated and efficient manner.

B. Coordinate installation of glazing with Section 08 80 00.

3.02 EXAMINATION

A. Verify that substrate and project conditions are suitable before beginning installation of frames.

B. Correct unsatisfactory condition before proceeding with installation.

C. Start of installation indicates acceptance of substrate and conditions.

3.03 INSTALLATION

A. Install frames plumb, level, rigid, and in true alignment as recommended in ANSI/SDI A250.11, NFPA 80, and in accordance with fire labeling requirement on fire-rated openings.

B. Screw jamb anchors securely to wall framing / structure using method recommended by manufacturer for permanently secure installation.

C. Coordinate installation of glazing stops by Section 08 80 00.

3.04 CLEAN

A. Clean and restore soiled surfaces. Remove scraps and debris, and leave site in a clean condition.

3.05 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

END OF SECTION

SECTION 08 80 00

GLASS AND GLAZING

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Fire-Rated Glass.

B. Glazing Accessories.

1.03 REFERENCES

A. References shall be the edition current as of the date of the Contract Documents.

B. ASTM C864 - Standard Specification for Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers.

C. GANA (GM) - GANA Glazing Manual.

D. GANA (SM) - GANA Sealant Manual.

E. UL (FRD) - Fire Resistance Directory.

1.04 SUBMITTALS

A. Refer to Division 01 for submittal procedures.

B. Product Data: Provide manufacturer's descriptive literature and performance data on each different type of glass and insulated glass unit and insulated composite metal panel unit specified.

1.05 PERFORMANCE REQUIREMENTS

A. Glass Thickness: Except where glass thickness is noted, select thickness of exterior glass to withstand dead loads and positive and negative live loads acting normal to plane of glass at design pressures calculated in accordance with requirements in the International Building Code and State / local codes.

1. Limit glass deflection to 1/200 or flexure limit of glass, whichever is less, with full recovery of glazing materials.

1.06 QUALITY ASSURANCE

A. Perform Work in accordance with GANA (GM) Glazing Manual and GANA (SM) Sealant Manual for glazing installation methods.

B. Installer Qualifications: Company specializing in the installation of products specified in this section on projects of similar scope and complexity, with a minimum of five (5) years of documented experience. Upon request, provided listing of projects completed within the last two (2) years along names and contact information of general contractors and building owner representative for each project.

1.07 ENVIRONMENTAL REQUIREMENTS

A. Do not install glazing when ambient temperature is less than 50 degrees F.

B. Maintain minimum ambient temperature before, during and 24 hours after installation of glazing compounds.

PART 2 - PRODUCTS

2.01 FIRE-RATED GLASS MATERIALS

A. Clear Fire-Rated Safety Laminated Glass:

1. Manufacturer / Product: Pilkington, *Pyrostop* is the basis of design and the standard of quality, function, performance and appearance required for this project.

    a. Pilkington; *Pyrostop* (specified, basis of design).

    b. Products by other manufacturers may be acceptable, subject to their ability to meet or exceed this standard and provide products conforming with the requirements of the Drawings and this Specification.

    c. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified product performance and appearance.

2. Thickness: 1-7/16-inch.

3. Fire Rating: 90 minute; UL (FRD) tested and listed.

4. Impact Safety Resistance: Comply with ANSI Z97.1 and CPSC 16 CFR 1201 Category 1 and II.

5. Permanently etch one (1) corner of each piece of glass indicating fire rating and Pyrostop logo; locate etch mark so it is visible after installation.

2.02 GLAZING ACCESSORIES

A. Setting Blocks: Neoprene, 80 to 90 Shore A durometer hardness, ASTM C864 Option I. Length of 0.1-inch for each square foot of glazing or minimum 4-inch x width of glazing rabbet space minus 1/16-inch x height to suit glazing method and pane weight and area.

B. Spacer Shims: Neoprene, 50 to 60 Shore A durometer hardness, ASTM C864 Option I. Minimum 3-inch long x one half the height of the glazing stop x thickness to suit application, self-adhesive on one face.

C. Glazing Tape: Closed cell polyvinyl chloride foam, coiled on release paper over adhesive on two sides, maximum water absorption by volume of 2 percent, designed for compression of 25 percent to effect an air barrier and vapor retarder seal.

1. Fire-Rated Glazing: Glazing tape shall conform to fire listing requirements of the fire-rated glazing.

2.03 FABRICATION

A. Fire-Rated Safety Laminated Glass: Fabricate using methods and equipment recommended by manufacturer.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the overall Work of this project in a coordinated and efficient manner.

B. Coordinate the installation of glazing in steel frames with Section 08 11 00.

3.02 EXAMINATION

A. Verify that openings for glazing are correctly sized and within tolerance.

B. Verify that surfaces of glazing channels or recesses are clean, free of obstructions that may impede moisture movement, weeps are clear, and ready to receive glazing.

C. Beginning of installation indicates acceptance of openings, substrate and conditions.

3.03 PREPARATION

A. Clean contact surfaces with solvent and wipe dry.

3.04 INSTALLATION – GENERAL

A. Install glazing in conformance with GANA (GM) Glazing Manual.

3.05 INSTALLATION – INTERIOR STEEL FRAMES – INTERIOR DRY METHOD (TAPE AND TAPE)

A. Cut glazing tape to length and set against permanent stops straight and true to line, do not project above sight line.

B. Place setting blocks at 1/4 points with edge block no more than 6-inches from corners.

C. Rest glazing on setting blocks and push against tape for full contact at perimeter of pane or unit.

D. Place glazing tape on free perimeter of glazing in same manner described above.

E. Install removable stop without displacement of tape. Exert pressure on tape for full continuous contact.

F. Knife trim protruding tape.

G. Workmanship: Glazing tape shall not extend beyond edge of stop or be installed crooked or wavy; remove and reglaze.

H. Fire-Rated Assemblies: Conform to fire rating requirements of glazing type and installation.

3.06 CLEANING

A. Remove glazing materials from finish surfaces.

B. Remove labels after Work is complete.

C. Clean glass and adjacent surfaces.

3.07 PROTECTION

A. Protect installed products until completion of project.

B. Repair or replace damaged products before Substantial Completion.

3.08 GLAZING TYPE SCHEDULE (REFER TO DRAWINGS FOR LOCATIONS)

A. **GL-1:** Single pane of 1-7/16-inch thick Fire-Rated Safety Laminated Glass (glass is fire-rated for 90 minutes, refer to Drawings for locations).

3.09 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices.

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

END OF SECTION

SECTION 09 22 16

NON-STRUCTURAL METAL FRAMING

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Non-Structural Metal Wall Framing.

1.03 PERFORMANCE REQUIREMENTS

A. Install metal framing in conformance with ASTM C754.

1.04 REFERENCES

A. References shall be the edition current as of the date of the Contract Documents.

B. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process.

C. ASTM A1003/A1003M - Standard Specification for Steel Sheet, Carbon, Metallic- and Nonmetallic-Coated for Cold-Formed Framing Members.

D. ASTM C645 - Standard Specification for Nonstructural Steel Framing Members.

E. ASTM C754 - Standard Specification for Installation of Steel Framing Members to Receive Screw-Attached Gypsum Panel Products.

F. ASTM C1513 - Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections.

1.05 SUBMITTALS

A. Refer to Division 01 for submittal procedures.

B. Product Data: Provide manufacturer’s data for each material specified in Part 2 including product criteria, load charts, and limitations.

1.06 QUALITY ASSURANCE

A. Perform Work in accordance with ASTM C754.

B. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section, with a minimum of five (5) years of experience.

C. Installer Qualifications: Experienced in performing work of this section with a minimum of five (5) years documented experience in installation of work of similar size, scope and complexity.

1. Installer shall employ only skilled, journeyman workers to install the work of this section.

PART 2 - PRODUCTS

2.01 NON-STRUCTURAL FRAMING

A. Non-Fire-Rated Assemblies: Use any of the products below at the locations noted on the Drawings.

B. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers may provide products:

1. Cemco.

2. ClarkDietrich Building Systems.

3. Craco Manufacturing, Inc.

4. Scafco, Steel Stud Company.

5. Steeler Construction Supply.

6. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements.

C. Framing Members, General: Comply with ASTM C645; roll-formed from hot-dipped galvanized steel; complying with ASTM A1003/A1003M and ASTM A653/A653M.

1. Base Metal Thickness:

a. 16 Gauge – 0.0538-inch minimum thickness.

b. 18 Gauge – 0.0428-inch minimum thickness.

c. 20 Gauge – 0.0329-inch minimum thickness.

D. Non-Structural Metal Studs:

1. Conform to ASTM C645, rolled galvanized steel, channel shaped, punched for utility access, width as indicated on Drawings.

a. Thickness: Minimum 20 gauge unless noted otherwise on the Drawings.

1) Provide double stud (one (1) 16 gauge and one (1) 20 gauge stud at each jamb of relite opening and at openings over 14-inches wide.

b. Channel Bridging: Cold-formed channel, 1-1/2 inches, 475 lbs. per 1,000 lineal feet, galvanized.

E. Tracks and Runners: Same material and thickness as studs.

F. Angle: Minimum 20 gauge, 2-inch x 2-inch or as required by condition.

G. Screw Fasteners for Framing: ASTM C1513, self-drilling, self-tapping screws of type intended for specific application.

H. Acoustic Sealant: Non-hardening, non-skinning, for use in conjunction with gypsum board.

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. Hilti; *CP 506 Smoke and Acoustic Sealant.*

b. Tremco; *Acoustical Sealant.*

c. U.S. Gypsum; *Sheetrock Brand Acoustical Sealant.*

d. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified products performance.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the Work of this project in a coordinated and efficient manner.

B. Coordinate framing installation with mechanical and electrical rough-in work located within framing.

C. Coordinate framing installation to accommodate requirements of finish materials.

D. Coordinate location and type of backing required by other trades.

E. Coordinate installation of framing and backing for support of GWB ends specified in Section 09 29 00.

F. Coordinate locations and size of additional studs or bracing required for work by other trades.

3.02 EXAMINATION

A. Prior to installation, inspect previous work of other trades. Verify that the Work is complete and accurate to the point where installation of non-structural metal framing may properly proceed.

3.03 LAYOUT

A. Lay out walls to dimensions and configuration shown on the Drawings; confirm the following:

1. Framing dimensions and layout shown on the Drawings fits within the building and provides the configuration and relationships shown, does not result in different size spaces or insufficient room for doors, windows, etc.

2. Do not start framing installation until conflicts and problems with work of other trades have been resolved.

B. Start of framing installation constitutes installers acceptance of responsibility for the following:

1. Building structure and site conditions are suitable for framing work to proceed as shown and specified.

2. Framing layout has been fully coordinated with work of other trades so that conflicts requiring rework or additional furring will not occur.

3.04 INSTALLATION – METAL FRAMING

A. Install framing plumb, straight and true in conformance with ASTM C754 and this section.

1. Size and spacing of wall framing shall be as indicated on Drawings.

B. Align and securely anchor top and bottom runners to building structural member as indicated on Drawings.

C. Fit runners under and above openings; secure intermediate studs at same spacing as wall studs. Increase gauge of runners as required to span across openings with deflection not exceeding L / 360.

D. Align stud web openings horizontally.

E. Secure studs to tracks using fastener method. Do not weld.

F. Stud splicing is not permitted.

G. Provide double stud as specified in Part 2 at each jamb of relite opening and at openings over 14-inches wide.

H. Coordinate installation of studs with requirements of relite frames or wall openings; install supports and attachments.

I. Openings: Frame openings required by other trades, coordinate location and sizes.

J. Provide single row of mid-height bridging minimum at walls, install additional rows of horizontal bridging in wall framing where required to maintain less than L / 240 deflection factor.

K. Tolerances:

1. Straight Alignment: No deviation allowed.

2. Out of Plumb: 1/8-inch in 12 feet maximum.

3. Variation of Surface Plane: 1/8-inch in 10 feet maximum.

3.05 REPAIR AND CLEAN-UP

A. Repair or replace damaged metal framing.

B. Remove from project site and legally dispose of construction debris associated with this work.

3.06 PROTECTION

A. Protect installed products and finished surfaces from damage during construction.

B. Do not overload framing during construction.

3.07 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices.

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

END OF SECTION

SECTION 09 29 00

GYPSUM BOARD ASSEMBLIES

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Interior Gypsum Board, Type X (GWB).

B. Cementitious Backer Board (CBB).

C. Finishing Gypsum Board.

D. Gypsum Board Assembly Cutting and Patching.

1.03 REFERENCES

A. References shall be the edition current as of the date of the Contract Documents.

B. ANSI A108.11 - American National Standard Specifications for Interior Installation of Cementitious Backer Units.

C. ANSI A118.9 - American National Standard Specifications for Test Methods and Specifications for Cementitious Backer Units.

D. ASTM C475/C475M - Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board.

E. ASTM C840 - Standard Specification for Application and Finishing of Gypsum Board.

F. ASTM C1002 - Standard Specification for Steel Self-Piercing Tapping Screws for Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs.

G. ASTM C1280 - Standard Specification for Application of Exterior Gypsum Panel Products for Use as Sheathing.

H. ASTM C1325 - Standard Specification for Fiber-Mat Reinforced Cementitious Backer Units.

I. ASTM C1396/C1396M - Standard Specification for Gypsum Board.

J. GA-214 - Levels of Finish for Gypsum Panel Products.

K. GA-216 - Application and Finishing of Gypsum Panel Products.

1.04 SUBMITTALS

A. Refer to Division 01 for submittal procedures.

B. Product Data: Submit manufacturer's product data for each proposed product sufficient to show compliance with each product specified.

1.05 QUALITY ASSURANCE

A. Applicator Qualifications: Company specializing in performing the work of this section with a minimum of five (5) years of consecutive, successful experience.

1.06 REGULATORY REQUIREMENTS

A. Conform to applicable codes and installation requirements for fire-rated assemblies referenced on the Drawings.

1.07 DELIVERY, STORAGE AND HANDLING

A. Deliver materials in original and unopened packages, containers, or bundles, with brand names and manufacturer's labels intact and legible.

B. Store materials in dry location, fully protected from weather and direct exposure to sunlight.

C. Stack gypsum board products flat and level, properly supported to prevent sagging or damage to ends and edges.

D. Store corner bead and other metal and plastic accessories to prevent bending, sagging, distortion, or other mechanical damage.

1.08 PROJECT CONDITIONS

A. Environmental Conditions: Establish and maintain environmental conditions for applying and finishing gypsum board to comply with ASTM C840 requirements, gypsum board manufacturer's recommendations, and the following; where requirements conflict, the most stringent requirement shall govern:

1. GWB Moisture Content: Do not begin taping and finishing of GWB until temperature of at least 50 degrees has been maintained for a period of 48 hours, and a GWB moisture content of 0.4% on a gypsum scale (12% on a wood scale) or less is achieved.

2. Lighting Level: Minimum of 15 candle power per square foot to be provided and maintained to assure quality workmanship.

3. Ventilation: Provide controlled ventilation during joint finishing operations, to eliminate excessive moisture. Avoid drafts during hot, dry weather to prevent finishing materials from drying too quickly.

PART 2 - PRODUCTS

2.01 GYPSUM BOARD MATERIALS

A. Manufacturers / Products:

1. Fire-Rated Assemblies:

a. Fire-Rated Assemblies with Proprietary Manufacturers / Products: Use the manufacturers / products required by the fire-rated assemblies referenced on the Drawings that list proprietary manufacturers / products whether specified herein or not.

b. Fire-Rated Assemblies with Non-Proprietary (Generic) Manufacturers / Products: Use any of the specified products below in fire-rated assemblies referenced on the Drawings that list generic GWB products.

2. Non-Fire-Rated Assemblies: Use any of the products below at the locations noted on the Drawings.

B. Interior Gypsum Board, Type X (GWB): Conform to ASTM C1396/C1396M.

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. CertainTeed; *Type X Gypsum Board*.

b. GP Gypsum; *ToughRock Fireguard X*.

c. National Gypsum; *Gold Bond Brand Fire-Shield X Gypsum Board*.

d. U.S. Gypsum; *Sheetrock Firecode X Gypsum Panels*.

2. Thickness: As noted on the Drawings.

3. Edges: Tapered.

4. Size: Largest size practicable to minimize joints in place and for least number of butt joints.

5. Ends: Square cut.

6. Location: Standard interior gypsum board used throughout the Project unless noted otherwise.

7. Fire-Rated: Type X, fire-rated, tested in accordance with ASTM E119 or UL 263.

C. Cementitious Backer Board (CBB): Conforms to ANSI A118.9, ANSI A108.11 and ASTM C1325 provide with manufacturer's standard edges.

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. U.S. Gypsum; *Durock Cement Board*.

2. Thickness: 5/8-inch.

3. Edges: Square.

4. Size: Largest size practicable to minimize joints in place.

5. Ends: Square cut.

6. Fire Rated: UL or WH tested and listed at fire rated walls noted on Drawings.

2.02 ACCESSORIES

A. Acoustic Sealant: Non-hardening, non-skinning, for use in conjunction with gypsum board.

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. Hilti; *CP 506 Smoke and Acoustic Sealant*.

b. Tremco; *Acoustical Sealant*.

c. U.S. Gypsum; *Sheetrock Brand Acoustical Sealant*.

d. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified products performance.

B. Outside Square Corners: Galvanized metal corner bead factory clad with paper tape.

1. Manufacturer / Product: Beadex, *Microbead* is the basis of design and the standard of quality, function, performance and appearance required for this project.

a. Beadex; *Microbead* (specified, basis of design).

b. Products by other manufacturers may be acceptable, subject to their ability to meet or exceed this standard and provide products conforming with the requirements of the Drawings and this Specification.

c. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified product performance and appearance.

C. J-Mold (Where GWB Abuts Dissimilar Material and Is Exposed to View): Galvanized metal J-shaped trim factory clad with paper.

1. Manufacturer / Product: Beadex, *B9J Tape-On "J" Trim* is the basis of design and the standard of quality, function, performance and appearance required for this project.

a. Beadex; *B9J Tape-On "J" Trim* (specified, basis of design).

b. Products by other manufacturers may be acceptable, subject to their ability to meet or exceed this standard and provide products conforming with the requirements of the Drawings and this Specification.

c. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified product performance and appearance.

D. Joint Materials: Provide products by manufacturer of gypsum board. Conform to ASTM C475/C475M and as recommended by gypsum board manufacturer for project conditions.

1. Interior Applications – Gypsum Board (except Water Resistant Panels): Ready-mixed vinyl-based joint compound:

a. Taping Compound: Type specifically formulated for embedding tape and accessories and for pre-filling.

b. Topping Compound: Type specifically formulated for finishing drywall over taping compound.

c. Joint Tape: Gypsum Board manufacturer's standard paper reinforcing tape.

2. Cementitious Backer Board:

a. Portland Cement Thin-Set Mortar:

1) Manufacturer / Product: U.S. Gypsum, *Durock Brand Latex-Modified Mortar Flexible Thin-Set High Performance* is the basis of design and the standard of quality, function, performance and appearance required for this project.

a) U.S. Gypsum; *Durock Brand Latex-Modified Mortar Flexible Thin-Set High Performance* (specified, basis of design).

b) Products by other manufacturers may be acceptable, subject to their ability to meet or exceed this standard and provide products conforming with the requirements of the Drawings and this Specification.

c) Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified product performance and appearance.

b. Joint Tape:

1) Manufacturer / Product: U.S. Gypsum, *Durock Brand Interior Joint Tape* is the basis of design and the standard of quality, function, performance and appearance required for this project.

a) U.S. Gypsum; *Durock Brand Interior Joint Tape* (specified, basis of design).

b) Products by other manufacturers may be acceptable, subject to their ability to meet or exceed this standard and provide products conforming with the requirements of the Drawings and this Specification.

c) Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified product performance and appearance.

E. Screws:

1. Interior Application: Conform to ASTM C1002; bugle-head steel, self-drilling type, black phosphate finish.

2. Cementitious Backer Board:

a. Manufacturer / Product: U.S. Gypsum, *Durock Brand Screws* are the basis of design and the standard of quality, function, performance and appearance required for this project.

1) U.S. Gypsum; *Durock Brand Screws* (specified, basis of design).

2) Products by other manufacturers may be acceptable, subject to their ability to meet or exceed this standard and provide products conforming with the requirements of the Drawings and this Specification.

3) Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified product performance and appearance.

F. Nails: Not allowed.

G. Sealer: Primer / Surfacer:

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. CertainTeed; *Level V Wall and Ceiling (Primer / Surfacer with M²Tech)*.

b. Hamilton *Drywall Products; Prep Coat (Level 5 Drywall Completion System)*.

c. U.S. Gypsum; *Primer-Surfacer Tuff-Hide*.

d. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified product performance and appearance.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the overall Work of this project in a coordinated and efficient manner.

B. Coordinate installation and requirements of fire-rated assemblies with other trades affected.

C. Coordinate location of backing for supporting ends of GWBwith Section 09 22 16.

D. Inspect finished surfaces with Section 09 90 00 painting applicator and project superintendent, mark areas that require additional finishing.

3.02 EXAMINATION

A. Confirm that the installation area(s) conform to the environmental conditions specified in Part 1 of this section.

B. Verify that project conditions are appropriate for work of this section to commence.

C. Confirm that the framing is straight, is within specified tolerances and meets minimum allowable deflection requirements.

D. Confirm that utility rough-in fits properly within framing width and will not prevent GWB from fitting tight to face of framing members.

E. Confirm that there is adequate temporary heat and light.

F. Do not start GWB installation until the conditions conform to requirements.

G. Beginning of installation indicates acceptance of framing and conditions.

3.03 FLOOR PROTECTION

A. Protect unsealed concrete floors from contact with GWB dust, taping mud and primer / surfacer using heavy paper or other method.

3.04 INSTALLATION – GYPSUM BOARD

A. Install GWB in conformance with ASTM C840, ASTM C1280, GA-216, and manufacturer's installation instructions.

1. Install in longest lengths possible for minimum number of joints.

2. Install to minimize butt end joints, especially in highly visible locations.

3. Comply with the installation requirements of fire-rated assemblies listed on the Drawings.

4. Use screws for attachment of all gypsum board.

B. Install full width panels with cut pieces only at top of wall (no "belly bands").

C. Place wrapped edges adjacent to one another; do not place cut edges or butt ends adjacent to wrapped edges.

D. Maintain 1/4-inch maximum gap between bottom of gypsum board and floor.

E. Single-Layer Installation: Install gypsum board, with edges and ends occurring over firm bearing.

F. Fire-Rated Assemblies Installation: Adhere to the installation requirements of the fire-rated assemblies referenced on the Drawings.

G. Acoustic Sealant: Install at perimeter of all sound walls in accordance with manufacturer's instructions and as follows:

1. Place continuous bead at perimeter of each layer of gypsum board.

2. Seal around all penetrations by conduit, pipe, ducts, rough-in boxes, and at other similar penetrations.

H. Run gypsum board full depth behind steel relite frames.

I. Dust and Debris: Remove GWB dust and debris from concealed spaces and from interior spaces inside of walls prior to covering.

3.05 INSTALLATION – CEMENTITIOUS BACKER BOARD (CBB)

A. Install CBB securely to wall framing members in conformance with manufacturer's installation instructions.

B. Attach units with screws at manufacturer's recommended spacing.

C. Fill joints with Portland Cement thin-set mortar and install reinforcing mesh joint tape centered on joints; trowel thin-set mortar flush and smooth.

D. Allow mortar to cure and thoroughly dry before installing fluid-applied coved base or solid surface panels.

3.06 INSTALLATION – TRIM AND ACCESSORIES

A. Corner Beads: Install at external corners in a single full length piece free of butt joints, using longest practical lengths, no short pieces; place into a solid bed of soft joint compound for secure installation.

1. Align bead straight and plumb.

2. Align juncture with other corner bead flush.

B. J-Shaped Edge Trim: Install at any exposed to view location where gypsum board abuts any dissimilar material or ends with exposed edge (around window frames, exposed structure, etc.).

1. Install heavy cardboard continuous at window perimeter to protect frame from dirt and damage.

3.07 GYPSUM BOARD LEVEL OF FINISH

A. Finish gypsum board (whether exposed to view or not) in accordance with GA-214 to the following minimum level of finish:

1. Painted Finish Exposed to View: Level 5 Smooth, substitute a coat of Primer / Surfacer (15 – 20 wet mil thickness) in lieu of skim coating with joint compound; sand surface of Primer / Surfacer smooth.

2. Surfaces Concealed from View with Thick Adhered Surface Finish (rubber base and stainless steel trim): Level 3, no texture.

3. Concealed from View without Surface Finish Above Suspended Lay-In Ceilings: Level 1.

B. Tape, fill and sand exposed joints, edges and corners to produce smooth surface ready to receive finishes.

1. Feather coats of joint compound so that camber is maximum 1/32-inch.

3.08 INSPECTION WITH PAINTER AND PROJECT SUPERINTENDENT

A. After taping and finishing is completed, coordinate an inspection walkthrough of all finished GWB surfaces with the painter and project superintendent using a 500 watt lamp aimed from side; mark any defects in the surface finish.

1. Repair defective areas not conforming to specified level of finish and seal repaired areas.

2. Do not proceed with application of Primer / Surfacer until defective, nonconforming areas have been refinished correctly.

3.09 APPLICATION – PRIMER / SURFACER

A. Apply Primer / Surfacer to surfaces exposed to view in accordance with manufacturer's installation instructions and at recommended application rate to achieve GA-214 Level 5 Smooth appearance, free of visible tape joint lines after finish painting is completed.

1. Spray-apply Primer / Surfacer to 15 - 20 mil wet film thickness applied in two (2) separate passes at 90 degrees to each other for proper coverage.

2. Sand surface of Primer / Surfacer lightly after it has dried to eliminate any unwanted stipple pattern or texture.

B. After application of Primer / Surfacer, carefully inspect walls with 500 watt lamp aimed from side and mark any defects in surface finish.

1. Fill / sand defective areas in surface finish and recoat with primer / surfacer.

3.10 CUTTING AND PATCHING – GYPSUM BOARD ASSEMBLIES

A. Coordinate with the work of other trades to minimize the cutting and patching work required and to avoid damaging existing finish surfaces not scheduled for replacement.

B. Cutting: Provide openings, cut holes, install corner bead or trim, etc. as required to accommodate the work of other trades and complete the work of this Project. Cut tight to items which penetrate gypsum board for minimum gap.

C. Patching: Edges of gypsum board patches shall be supported on wood framing or wood backing, cut existing gypsum board back to framing members where openings are required. Finish as specified herein and blend in with existing surface so that patched areas are not visible and match adjacent finish. Install acoustic sealant at perimeter of penetrations and finish tape tight to eliminate gaps and openings.

D. Fire-Rated Assemblies: Maintain fire-rated integrity of existing wall and ceiling assemblies.

3.11 TOLERANCES

A. Gap between Bottom of GWB and Floor: 1/4-inch maximum.

B. Maximum Variation of Finished Gypsum Board Surface from True Flatness: 1/8-inch in 10 feet in any direction.

C. Butt Joint Finishing: Not readily visible under the normal lighting conditions found for any given area / surface.

D. Finishing Tolerances: All exposed surfaces shall be smooth and free from visible ridges, waves, ripples, holes, defects, delamination, roughness, depressions, screw pops, etc. Taped joints shall not be visible after finish paint application.

3.12 CLEAN UP

A. Remove excess gypsum board and finishing materials from the site.

B. Remove gypsum board scraps and dust from all concealed spaces including interior spaces of wall framing.

C. Remove gypsum dust, taping mud and primer / surfacer completely from window frames, door frames, subfloor surfaces and any surface / material exposed to view.

1. Subfloor cleanliness / condition shall conform to floor covering installation requirements.

3.13 WORKMANSHIP

A. Gypsum wallboard shall be installed and finished using the best workmanship, including the following:

1. No damaged board or paper face.

2. Ends centered on framing.

3. GWB tucked full depth behind hollow metal door frames.

4. Gap at bottom of GWB 1/4-inch or less.

5. Cut-outs for outlets and devices cut neatly with saw or router.

6. GWB fastened tight to face of studs to eliminate screw pops.

7. Acoustic sealant consistently applied to all openings and perimeters.

8. All screws that do not engage framing removed.

9. Taped joints full bedded in taping compound and free of air pockets.

10. Butt joints finished with minimal thickness and tapered out for flat appearance.

11. Taped joints smooth and flat so as to disappear after painting.

12. Paper face not roughened by sanding.

13. Bottom of GWB behind rubber base properly finish smooth and free of roughness.

B. Gypsum wallboard installed and finished with improper or poor workmanship shall be removed and replaced at Contractor's expense.

3.14 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices.

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

2. Fire-Rated Assemblies: Contractor is responsible to verify that the installation and materials conform to the requirements of the fire-rated assemblies referenced on the Drawings.

B. Schedule of Required Inspections by Contractor; confirm installation and workmanship are as shown / specified:

1. Inspect framing for conformance to specified surface tolerances.

2. Inspect GWB installation.

3. Inspect trim installation.

4. Inspect taping and finish application.

5. Inspect finish on bottom of walls with rubber base.

6. Inspect finished GWB surfaces after sealer is applied with painter and GWB finish applicator to identify any finishing defects requiring correction prior to finish application and painting.

END OF SECTION

SECTION 09 30 00

TILE

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Tile Work.

1. Existing Granite Wall Panel Tile.

2. Terrazzo Floor Tile.

1.03 REFERENCES

A. References shall be the edition current as of the date of the Contract Documents.

B. ANSI A108.1a - American National Standard Specifications for Installation of Ceramic Tile in the Wet-Set Method, with Portland Cement Mortar.

C. ANSI A108.10 - American National Standard Specifications for Installation of Grout in Tilework.

D. ANSI A137.1 - American National Standard Specifications for Ceramic Tile.

E. TCNA (HB) - Handbook for Ceramic, Glass, and Stone Tile Installation.

1.04 SUBMITTALS

A. Refer to Division 01 for submittal procedures.

B. Product Data: Submit manufacturer’s product data for each different material specified.

1.05 QUALITY ASSURANCE

A. Conform to ANSI A137.1.

B. Conform to TCNA (HB) Handbook for Ceramic, Glass, and Stone Tile Installation.

C. Conform to manufacturer's installation instruction.

1.06 QUALIFICATIONS

A. Manufacturer Qualifications: Company specializing in the manufacture of products specified in this section with a minimum of ten (10) years of documented experience.

B. Installer Qualifications: Company specializing in applying the work of this section with a minimum of five (5) years of documented experience and approved by product manufacturer.

1.07 DELIVERY, STORAGE AND HANDLING

A. Deliver, store and protect products as recommended by the manufacturer.

1.08 ENVIRONMENTAL REQUIREMENTS

A. Maintain 50 degrees F. minimum during installation of mortar materials.

PART 2 - PRODUCTS

2.01 WALL TILE

A. Existing Granite Wall Panel Tile: Existing Granite Wall Panel Tiles shall be carefully removed from the existing wall during selective demolition and salvaged for reuse. Once removed, turn the panels over to the Owner for storage. The Owner will return the panels to the contractor when they are needed to be reinstalled. Contractor shall provide all required setting materials.

2.02 FLOOR TILE

A. Terrazzo Floor Tile: Owner has extra stock of the required terrazzo floor tiles for the project. Contractor is to coordinate with the Owner to obtain the tiles. The tiles are an Owner-Furnished, Contractor-Installed material. Contractor shall provide all required setting materials.

2.03 CEMENT BACKER BOARD

A. Cement Backer Board (CBB): Installation and joint treatment is specified in Section 09 29 00.

2.04 SETTING / INSTALLATION SYSTEMS (MORTAR, GROUT, AND ANTI-FRACTURE MEMBRANE)

A. General Requirements:

1. Manufacturer / Product Selection: Mortar and grout products installed shall be manufactured by a single manufacturer selected from any of those listed below (mixing different manufacturer’s products is not allowed).

B. Mortar Bond Coat Materials (Thinset):

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. Custom Building Products; *Prolite Tile and Stone Mortar.*

b. Custom Building Products; *MegaLite Crack Prevention Mortar.*

c. Laticrete; *255 MultiMax.*

d. Laticrete; *254 Platinum.*

e. Mapei; *Kerabond+Keralastic* (for floor and wall applications).

f. Mapei; *Ultracontact* (for floor applications only).

g. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified products performance and colors.

C. Epoxy Grout: Select any of the following manufacturer's products:

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. Custom Building Products; *CEG-Lite 100% Solids Commercial Epoxy Grout.*

b. Laticrete; *SpectraLOCK PRO Grout.*

c. Mapei; *Kerapoxy*.

d. Substitutions: Refer to Division 00 and Division 0101 60 00 for substitution procedures and requirements. Proposed substitutions must match specified products performance and colors.

2. Color: Match existing grout colors.

D. Anti-Fracture Membrane:

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. Custom Building Products; *Crack Buster Pro Crack Prevention Mat Underlayment.*

b. Laticrete; *Blue 92 Anti-Fracture Membrane.*

c. Mapei; *Mapelastic SM.*

d. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified products performance.

2. Provide width as recommended by manufacturer for size of tile; provide manufacturer's recommended primer.

2.05 ACCESSORIES

A. Provide fillers, admixtures, and adhesives as required to suit conditions for complete installation.

B. Sealant: High performance, single component neutral cure silicone sealant; in color matching grout joint; provide backer rod and bond breaker tape of type recommended by sealant manufacturer.

1. Manufacturers / Products: Subject to their ability to conform to the requirements of the Drawings and this Specification, the following manufacturers / products may be provided:

a. Custom Building Products; *Commercial 100% Silicone Sealant.*

b. Laticrete; *Latasil.*

c. Mapei; *Mapesil T.*

d. Substitutions: Refer to Division 00 and Division 01 for substitution procedures and requirements. Proposed substitutions must match specified products performance and colors.

2. Primer: Manufacturer's recommended primer for the sealant that is used.

C. Metal Trim / Edge Strips: Provide metal trim made to fit flooring conditions:

1. Terrazzo Floor Tiles: Provide metal trim pieces between tiles to match the existing installation condition in the area where the tiles are to be installed.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the overall Work of this project in a coordinated and efficient manner.

B. Coordinate installation to follow dry out of building and completion of GWB.

3.02 EXAMINATION

A. Inspect substrate surfaces and site conditions and verify that they conform to manufacturer's requirements and are acceptable to receive work.

B. Inspect cement backer board installed by Section 09 29 00.

C. Identify any unacceptable cracks requiring filling or repair.

D. Confirm that subfloor surface tolerances are within 1/4-inch in 10 feet and 1/16-inch in 1 foot per TCNA (HB) recommendations.

E. Measure spaces / rooms for square and parallel walls.

F. Confirm that substrates are not contaminated with any material / substance that would impair the adhesion of the tile to the substrate.

G. Do not proceed with installation until substrate surfaces and site conditions conform to all requirements.

H. Beginning of installation indicates installer accepts substrate surfaces and site conditions as conforming to all requirements.

3.03 PREPARATION

A. Prepare substrate surfaces and conditions for application as required by manufacturer of thin-set bond coat and waterproofing membrane.

B. Subfloor Cracks:

1. Install anti-fracture membrane over cracks in subfloor in accordance with manufacturer's installation instructions.

2. Seal substrate surface cracks with filler as recommended by waterproofing membrane manufacturer.

C. Thoroughly remove any substances on substrate surfaces that would negatively affect the installation or bond strength of the tile.

D. Protect surrounding work from damage or disfiguration.

E. Vacuum clean existing surfaces and damp clean.

3.04 CEMENT BACKER BOARD – JOINT TREATMENT

A. Cement Backer Board Installation: Specified in Section 09 29 00.

B. Joint Treatment: Specified in Section 09 29 00.

3.05 MIXING – THINSET MORTAR AND GROUT

A. Mix thinset mortar and grout in strict conformance with manufacturer's mixing instructions using specified products.

1. Machine Mixing: Mixer shall be a rotating blade mortar mixer. Do not over mix. Stop mixer and dump mortar from mixer promptly.

2. Hand Mixing: Pre-mix the dry ingredients (sand and cement). Place liquid in clean container or mixing box, add dry materials and mix. Adjust amount of liquid or dry material to obtain proper consistency.

3.06 TILE LAYOUT AND CUTTING

A. Tile Layout:

1. General: Avoid narrow cuts, small pieces or rows that taper in width; consult with Architect to determine best resolution before proceeding with cutting and installation.

2. Floors: Lay out tile in each room for equal borders and uniform appearance; if layout results in tile cuts less than ½ the tile width, consult with Architect and adjust layout as directed.

3. Walls: Lay out tile to avoid cutting tile horizontally; vertical cuts shall be at inside corners. When size of wall and floor tile match, align joints.

B. Cut tile to fit the spaces / rooms and configuration shown.

1. Cuts shall be accurate and align from tile to tile in a uniform line.

2. Cuts shall allow for joint width matching width of field tile.

C. Cut and fit tile to penetrations through tile, form corners and bases neatly.

3.07 TILE PATTERNS

A. Match the existing wall and floor tile patterns.

3.08 INSTALLATION – GENERAL

A. Install tile and grout in accordance with applicable requirements of ANSI A108.1a through ANSI A108.10, manufacturer's instructions, and TCNA (HB) Handbook recommendations.

B. Lay tile to pattern indicated. Do not interrupt tile pattern through openings.

C. Place metal trim / edge strips at exposed tile edges.

D. Place tile joints uniform in width, subject to variance in tolerance allowed in tile size. Make joints watertight, without voids, cracks, excess mortar, or excess grout.

E. Set tile firmly in place fully bonded to setting bed and with top surface flush with adjacent tile; align tile square and with straight parallel grout lines of uniform width.

1. Lippage of tile edges is not acceptable.

F. Form internal angles coved and external angles bullnosed, or as shown on Drawings.

G. Joint Patterns: Lay out tile according to patterns indicated on Drawings, or if not shown, in a grid pattern with floor joints aligning with wall and trim joints. Install joints straight and of uniform width.

H. Mortar shall be applied with notched trowel using scraping motion to work the material into good contact with the surface to be covered.

1. Trowel notches shall conform to size / spacing recommended by thinset mortar manufacturer.

I. Only as much mortar shall be applied as can be covered within 20 to 30 minutes, or is still tacky. Tile shall be aligned to show uniform joints and then allowed to set firm.

J. Back-butter large tile with mortar to assure full coverage and bond over entire tile surface.

K. Install tile under or behind equipment or fixtures.

L. Sound tile after setting. Replace hollow sounding units.

M. Keep control joints free of mortar or grout. Apply sealant to joints.

N. Allow tile to set for a minimum of 48 hours prior to grouting.

O. Grout tile joints working grout completely into full depth of joints.

P. Clean excess grout from exposed tile surfaces.

Q. Include sealant-filled joint at floor perimeters and wall inside corners and where system abuts different material and substrates.

1. Sealant color shall match grout.

2. Provide backer rod or bond breaker tape compatible with sealant.

3. Install backer rod or bond breaker and sealant in accordance with manufacturer’s installation instructions.

R. Apply sealant-filled joints at junction of tile and dissimilar materials, at junction of dissimilar planes and wherever else required to prevent cracks. Match grout color.

3.09 INSTALLATION SCHEDULE

A. General: All tile work shall be performed in conformance with installation methods outlined in the Tile Council of North America (TCNA (HB)) Handbook For Ceramic, Glass, and Stone Tile Installation as follows:

B. Floor Tile Over Concrete Slab on Grade: Thin-set application per TCNA (HB) Installation Method F113, using thin-set mortar bond coat over concrete slab on grade.

C. Wall Tile Over Cement Backer Board: Thin-set application per TCNA (HB) Installation Method W244C using latex thin-set mortar bond coat over Cement Backer Board.

1. Grout: Epoxy Grout.

D. Expansion Joints (Vertical and Horizontal): Install sealant filled joints per TCNA (HB) Installation Method EJ171 at perimeter of floors and inside corners of walls, and where shown on Drawings; match sealant color to grout color.

3.10 CLEANING

A. Clean tile surfaces; remove grout haze and dirt.

3.11 PROTECTION

A. Protect finished installation from damage.

B. Do not permit traffic over finished floor surface.

3.12 WORKMANSHIP

A. Tile shall be installed and finished using the best workmanship, including the following:

1. No cut tile “slivers” less than one-half standard tile width.

2. Floor tile surface installed in a flat plane sloping to drain and free of low or high spots.

3. Tile edges aligned flush and free of lippage.

4. All tile fully bonded / adhered to substrate.

5. Straight grout lines with uniform width.

6. Grout full to top edge of tile and not recessed down in joint.

7. No voids or open pockets in grout joints.

8. Grout haze completely removed from tile surfaces.

B. Tile installed with improper or poor workmanship shall be removed and replaced at Contractor's expense.

3.13 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices.

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

END OF SECTION

SECTION 10 26 00

WALL PROTECTION

PART 1 - GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general conditions of the Contract, including Divisions 00 and 01 Specification Sections, apply to work of this section.

1.02 SECTION INCLUDES

A. Wall Protection:

1. Custom Stainless Steel Wall Protection around Wall Opening.

1.03 REFERENCES

A. References shall be the edition current as of the date of the Contract Documents.

B. ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications.

1.04 SUBMITTALS

A. Refer to Division 01 for submittal procedures.

B. Product Data: Provide product data for each specified product, including installation details / instructions.

C. Shop Drawing: Provide shop drawing for wall protection showing the location of each type of wall protection on a floor plan.

D. Samples: Provide two (2) samples of each specified finish for the wall protection.

1.05 PROJECT CONDITIONS

A. Materials must be acclimated in an environment of 65 to 75 degrees F for at least 24 hours prior to beginning installation.

PART 2 - PRODUCTS

2.01 CUSTOM STAINLESS STEEL WALL PROTECTION

A. Custom Stainless Steel Wall Protection:

1. Materials:

a. Stainless Steel Sheet: ASTM A240/A240M, 16 gauge, type 304 or 316 stainless steel, with AISI No. 4 brushed finish

b. Mounting Adhesive: As recommended by the manufacturer for adhering stainless steel sheets to the substrates indicated on the Drawings.

2. Fabrication:

a. Opening Surround: Each piece shall be one-piece construction. Each inside corner shall be constructed with a 45 degree mitre on the face of the opening. Design intent is to match the appearance and installation of the existing stainless steel opening surrounds in the building. Adhere to substrate using adhesive recommended by the manufacturer.

PART 3 - EXECUTION

3.01 COORDINATION

A. Review, coordinate and accommodate work of other trades that interface with, affect or are affected by the work of this section so as to facilitate the execution of the overall Work of this project in a coordinated and efficient manner.

B. Coordinate location, layout and type of backing in walls to receive accessories with Section 09 22 16.

C. Coordinate installation to follow installation of interior finishes and final painting of walls.

3.02 EXAMINATION

A. Verify backing and installation conditions are correct before starting work.

B. Start of installation indicates acceptance of backing and installation conditions.

3.03 INSTALLATION – CUSTOM STAINLESS STEEL WALL PROTECTION

A. Install stainless steel wall protection securely to wall surfaces using adhesive in locations shown on the Drawings.

B. Fit tight to wall free of open gaps / cracks at edges.

C. Provide temporary brace to hold wall protection in place until adhesive sets up.

3.04 FIELD QUALITY CONTROL

A. Contractor Quality Control: Employ / assign quality control personnel to monitor the work of this section for conformance to the requirements of the Contract Documents and to good construction practices.

1. Contractor is solely responsible for managing and controlling the quality of the work and conformance with the requirements of the Contract Documents.

END OF SECTION